TypeScript vs WebAssembly Performance Benchmark

Comprehensive benchmark suite for comparing TypeScript and WebAssembly implementations of data structures and algorithms

Benchmark Configuration

Number of Iterations: 100 (higher numbers provide more accurate results but take longer)

Benchmark Tests:

☑ **Sorting** ☑ **AVL Tree** ☑ **Graph**

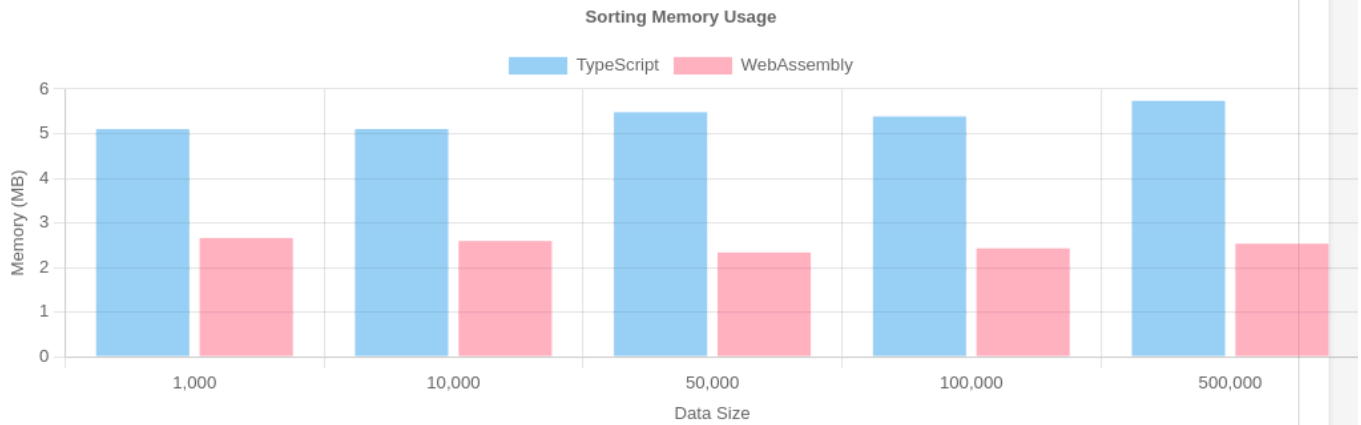
Start Comprehensive Benchmark | Download Results (CSV)

Sorting Algorithm

Data Sizes:

☑ **1,000** ☑ **10,000** ☑ **50,000** ☑ **100,000** ☑ **500,000**

100%

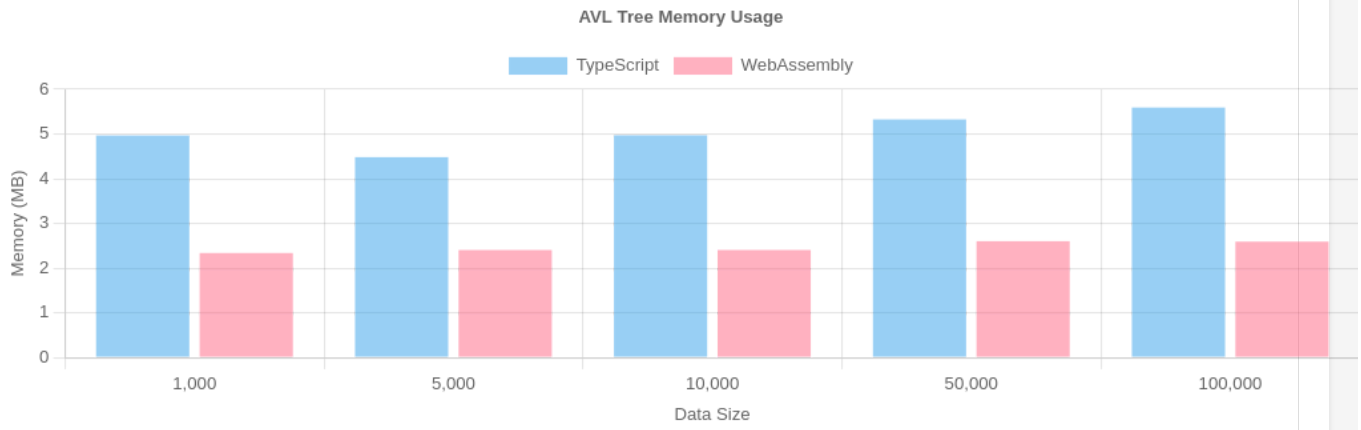


AVL Tree Operations

Data Sizes:

☑ **1,000** ☑ **5,000** ☑ **10,000** ☑ **50,000** ☑ **100,000**

100%

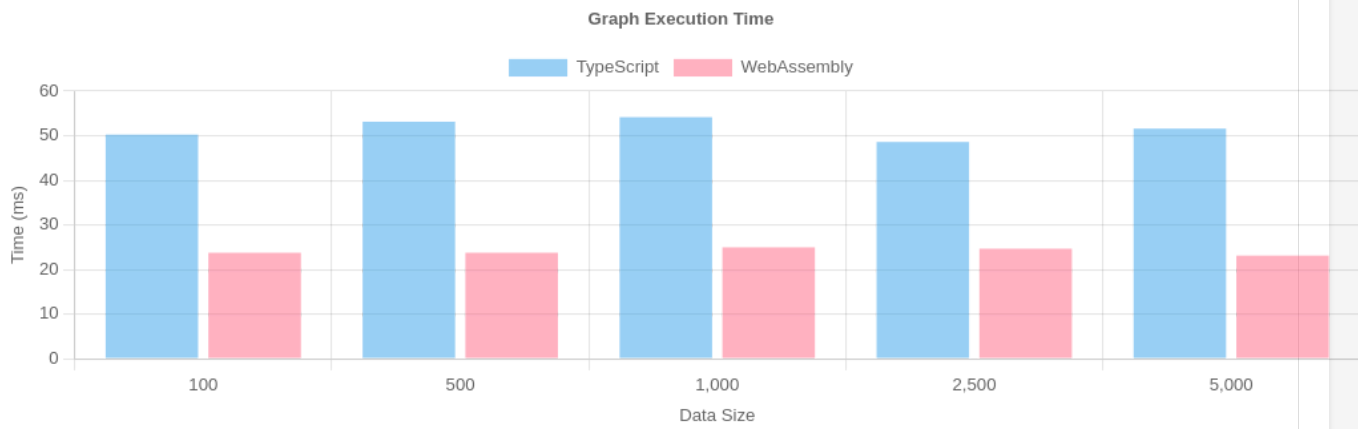


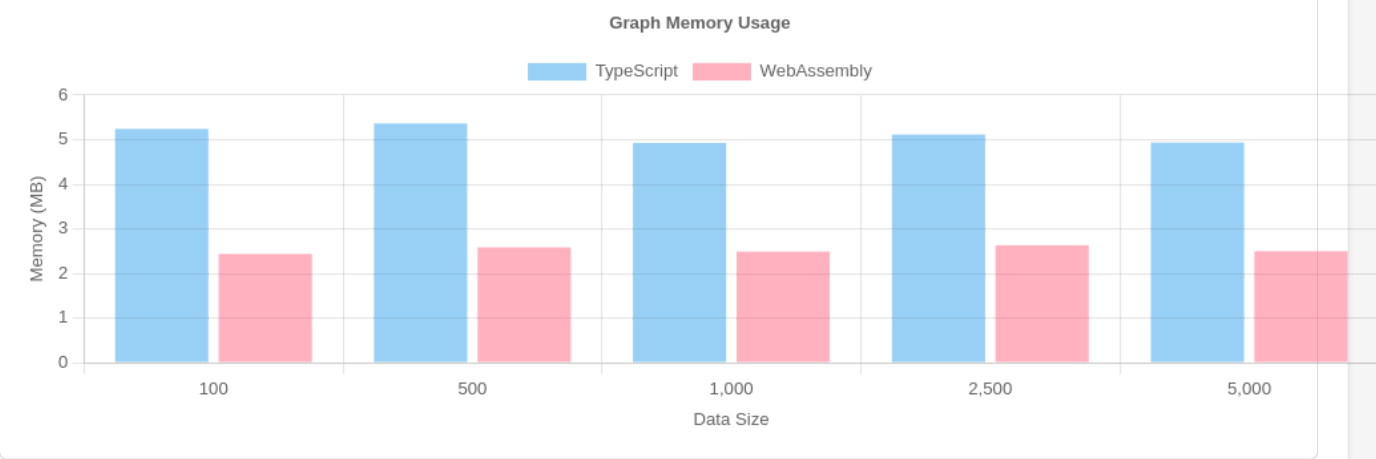
Graph Algorithm (Dijkstra)

Node Counts:

☑ **100** ☑ **500** ☑ **1,000** ☑ **2,500** ☑ **5,000**

100%





Benchmark Results

Summary

Sorting

Size: 1,000 - Wasm is **2.00x** faster (TS: 51.95ms, Wasm: 25.96ms)
Size: 10,000 - Wasm is **1.98x** faster (TS: 47.50ms, Wasm: 23.97ms)
Size: 50,000 - Wasm is **2.07x** faster (TS: 49.38ms, Wasm: 23.80ms)
Size: 100,000 - Wasm is **1.98x** faster (TS: 53.13ms, Wasm: 26.85ms)
Size: 500,000 - Wasm is **1.98x** faster (TS: 54.16ms, Wasm: 27.42ms)

AVL Tree

Size: 1,000 - Wasm is **2.18x** faster (TS: 52.39ms, Wasm: 24.01ms)
Size: 5,000 - Wasm is **2.33x** faster (TS: 54.51ms, Wasm: 23.37ms)
Size: 10,000 - Wasm is **1.88x** faster (TS: 44.94ms, Wasm: 23.95ms)
Size: 50,000 - Wasm is **2.03x** faster (TS: 49.30ms, Wasm: 24.23ms)

```
[3:33:14 AM] Completed Graph test (size: 5,000, iteration: 32/100)
[3:33:14 AM] Completed Graph test (size: 5,000, iteration: 33/100)
[3:33:14 AM] Completed Graph test (size: 5,000, iteration: 34/100)
[3:33:15 AM] Completed Graph test (size: 5,000, iteration: 35/100)
[3:33:15 AM] Completed Graph test (size: 5,000, iteration: 36/100)
[3:33:15 AM] Completed Graph test (size: 5,000, iteration: 37/100)
[3:33:15 AM] Completed Graph test (size: 5,000, iteration: 38/100)
[3:33:15 AM] Completed Graph test (size: 5,000, iteration: 39/100)
[3:33:15 AM] Completed Graph test (size: 5,000, iteration: 40/100)
```